

FARNBOROUGH, United Kingdom

WMO: 037680

Lat: 51.2800N

Lon: 0.7725W

Elev: 213

StdP: 14.58

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.4	26.2	20.8	15.8	28.5	23.1	17.6	30.5	27.8	50.7	25.3	50.6	1.9	230	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	81.3	66.1	77.5	64.6	74.3	63.0	68.5	77.0	66.4	74.4	64.6	71.7	8.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	95.3	71.7	63.5	88.4	69.4	61.9	83.4	67.6	32.9	76.6	31.1	74.4	29.8	71.8	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.7	20.0	17.8	DB	17.7	86.8	3.0	3.9	15.5	89.6	13.8	91.9	12.1	94.1	9.9	96.9	(4)
			WB	17.3	72.9	3.5	4.0	14.7	75.7	12.7	78.0	10.7	80.2	8.1	83.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	40.5	41.0	44.0	48.4	53.9	59.5	63.5	62.7	57.7	52.2	45.6	41.4	(6)
(7)		DBStd	9.88	6.36	6.05	5.56	5.50	5.32	4.57	4.48	4.49	4.51	5.55	6.13	6.94	(7)
(8)		HDD50	1332	298	255	197	91	21	0	0	0	2	41	154	273	(8)
(9)		HDD65	5213	760	672	652	497	345	172	84	101	222	397	581	730	(9)
(10)		CDD50	1673	2	3	10	43	142	287	419	393	233	109	24	8	(10)
(11)		CDD65	79	0	0	0	0	1	9	37	28	3	1	0	0	(11)
(12)		CDH74	793	0	0	0	3	24	123	359	250	31	3	0	0	(12)
(13)		CDH80	157	0	0	0	0	0	20	77	59	1	0	0	0	(13)
(14)	Wind	WSAvg	8.1	9.3	9.1	8.9	8.1	8.4	7.9	7.5	7.2	7.1	7.7	7.7	8.6	(14)
(15)	Precipitation	PrecAvg	27.9	3.0	2.1	1.7	2.1	1.9	2.0	1.9	2.3	2.2	2.9	3.1	2.7	(15)
(16)		PrecMax	37.4	7.2	5.0	4.4	5.6	4.1	5.0	4.8	4.5	4.6	6.1	6.3	5.1	(16)
(17)		PrecMin	20.6	0.4	0.2	0.3	0.1	0.2	0.5	0.5	0.4	0.4	1.1	1.2	0.5	(17)
(18)		PrecStd	4.5	1.5	1.3	0.9	1.3	1.0	1.2	0.9	1.3	1.1	1.3	1.5	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	57.2	63.9	72.7	76.9	82.6	86.1	86.0	78.0	69.2	60.8	57.0	(19)
MCWB			53.0	51.3	52.7	57.8	61.4	65.6	67.8	68.1	65.5	61.8	58.0	54.1	(20)	
2%		DB	53.6	53.9	58.7	66.4	72.9	77.4	81.9	80.6	73.3	64.7	58.4	55.4	(21)	
		MCWB	51.2	50.5	50.7	55.1	60.4	63.8	66.2	66.4	63.7	59.2	56.0	53.2	(22)	
5%		DB	51.8	52.0	55.6	62.0	68.8	73.4	78.0	75.9	69.5	62.5	56.7	53.9	(23)	
		MCWB	49.5	48.9	49.8	52.8	58.4	62.0	65.0	64.6	61.4	58.2	54.5	51.8	(24)	
10%		DB	50.1	50.2	53.2	58.5	64.4	70.1	74.2	72.1	66.7	60.7	55.0	52.0	(25)	
		MCWB	47.8	47.4	48.6	50.9	56.0	60.3	63.4	62.8	60.0	57.2	52.7	49.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.6	53.5	55.1	60.0	65.0	69.2	71.8	72.2	68.2	63.0	58.6	55.0	(27)
MCD B			54.8	55.6	59.9	68.5	72.4	78.4	79.0	79.8	73.9	67.3	59.9	56.0	(28)	
2%		WB	51.8	51.2	52.6	56.4	61.7	65.7	68.3	68.5	65.3	60.9	56.6	53.6	(29)	
		MCD B	53.3	53.1	56.2	64.5	69.8	74.6	77.8	76.3	70.7	63.5	57.8	55.0	(30)	
5%		WB	49.9	49.4	51.1	53.9	59.3	63.2	66.2	66.1	63.0	59.2	54.8	52.2	(31)	
		MCD B	51.5	51.4	54.4	59.5	66.6	70.9	75.3	72.9	67.4	61.6	56.3	53.7	(32)	
10%		WB	48.2	47.7	49.5	52.1	57.2	61.4	64.4	64.2	61.2	57.8	53.0	50.1	(33)	
		MCD B	49.9	49.9	52.7	57.3	63.4	68.4	72.3	70.3	65.1	60.0	54.6	51.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.7	11.9	14.5	16.8	16.9	17.2	17.6	17.2	16.9	14.1	11.8	10.4	(35)
MCD BR			10.5	13.0	18.4	23.0	23.9	24.3	25.1	23.9	21.1	14.6	10.8	9.3	(36)	
MCWBR			10.2	10.2	12.5	13.6	13.2	13.3	12.7	12.2	12.9	10.1	9.5	9.1	(37)	
5% WB		MCD BR	11.0	11.6	15.4	19.8	21.1	20.8	21.4	19.9	18.1	12.7	10.9	9.5	(38)	
		MCWBR	11.0	10.1	11.3	12.8	13.0	12.9	12.5	12.5	12.9	10.0	10.3	9.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.345	0.382	0.403	0.406	0.405	0.399	0.391	0.375	0.358	0.332	0.319	(40)	
taud		2.366	2.373	2.266	2.227	2.259	2.289	2.312	2.354	2.393	2.419	2.410	2.354	(41)		
Ebn at Noon		215	246	256	265	269	270	270	266	257	238	213	198	(42)		
Edh at Noon		20	26	35	40	41	40	38	35	30	24	19	17	(43)		
(44)	All-Sky Solar Radiation	RadAvg	260	473	828	1287	1558	1724	1634	1361	1021	588	312	205	(44)	
(45)		RadStd	21	61	111	141	97	172	151	108	84	51	35	24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page